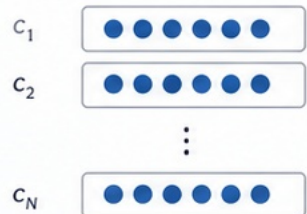


Candidates

Candidate concept embeddings



Token + Position
Embedding

Transformer Encoder $\times L$

Add & Norm

Feed-Forward
Network

Add & Norm

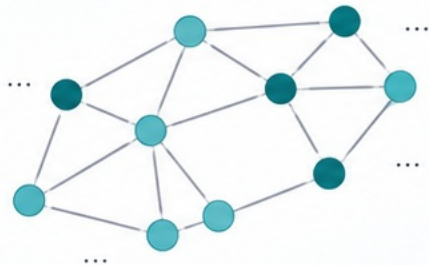
Multi-Head
Self-Attention

Candidate representations

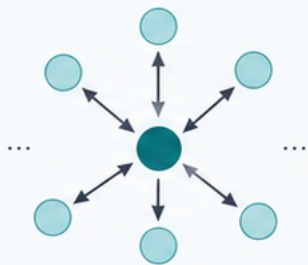


Ontology graph embedding

Ontology graph



Graph Encoder (Message Passing) $\times T$

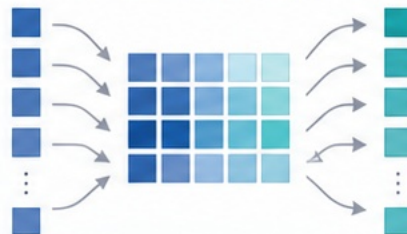


Graph embeddings

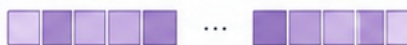


Risk prediction & gating

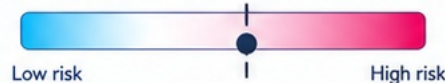
Cross-attention
(candidates $\leftrightarrow$ graph)



Contextualized candidate features



Risk score

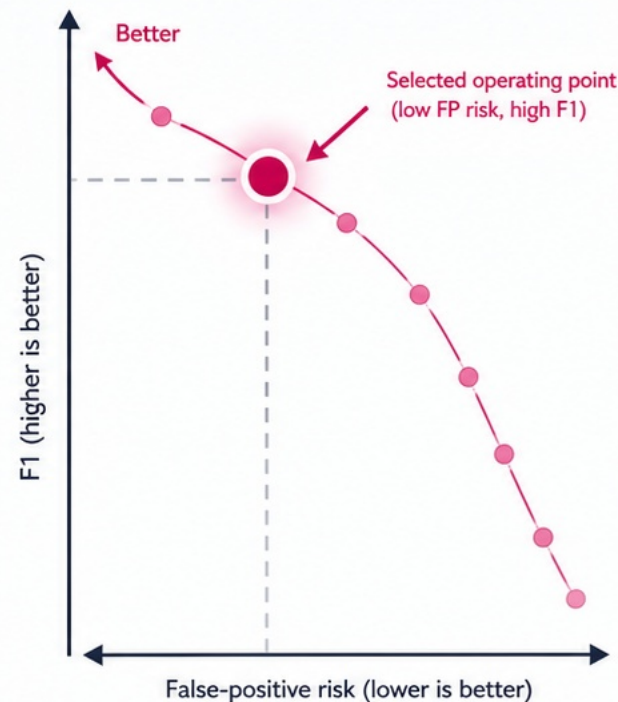


Calibrated risk gate
(adaptive threshold)



Frontier

Risk-F1 Pareto frontier



Accept



High confidence
Low risk

Defer



Low confidence
Higher risk